

Comparative Study of Machine Learning Algorithms for Voice based Gender Identification

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Abstract—This study presents an approach to the recognition of speech signal using frequency spectral information with Mel frequency for the improvement of speech, and proposes a novel dialect-aware ASR system that enhances speech recognition while simultaneously classifying Telugu dialects. MFCC coefficients capture spectral features of the voice, and have been used for emotion recognition, Mel-Frequency Cepstral Coefficients (MFCCs) represent the unique speech information in the frequency domain.

All audio samples are provided at a sampling rate of 16 kHz, and all audio recordings were provided in FLAC format and subsequently converted to NumPy arrays with a consistent sampling rate of 16 kHz. Wav2Vec2 is a self-supervised model initially used for human speech recognition tasks, wav2vec (2019) showed that raw audio can be pretrained using a contrastive objective to learn contextual speech representations. Wav2vec 2.0 enables us to build better speech recognition systems for many more languages and domains with much less annotated data, and wav2vec (2019) showed that raw audio can be pretrained using a contrastive objective to learn contextual speech representations. In contrast, pooling operations reduce variable-length sequences to fixed dimensions through mathematical operations that aggregate information, and feature embedding is the process of transforming data into continuous vector representations.

As ML algorithms, Support Vector Machine (SVM), Naive Bayes (NB), Random Forest (RF), Decision Tree (DT), K-Nearest Neighbor (KNN), Logistic Regression (LR) were used for classification tasks. Data augmentation techniques, such as waveform, noise adding, pitch shifting, and time stretching, are applied to improve the generalization capability of the model, and data augmentation offers a promising approach to mitigate class imbalance by artificially increasing the size of the minority class.

Wav2Vec 1.0, while generally outperformed by Wav2Vec 2.0, still provided a more efficient alternative with competitive performance, and the results show that combining all three deep embeddings achieves an AUROC of 89.49% and accuracy. This study presents a deep learning framework for speaker diarization, and this work explores the effectiveness of CAE with SSL-based speech representations to obtain improved AWEs, and Whisper large embeddings paired with a LogR classifier demonstrated an impressive average accuracy. This study highlights the effectiveness of self-supervised learning strategies for tasks such as automatic speech recognition, and a deep learning architecture that merges attentional and residual convolutional networks for age and gender classification.

Index Terms—MFCC, pitch, audio processing, Wav2Vec2, Random Forest, SVM, MLP, XGBoost, transgender voice recognition

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I. INTRODUCTION

Automatic gender classification based on voice signals plays a crucial role in biometric authentication, speech analytics,

and human-computer interaction. Developers use algorithmic gender classification in human-computer interaction and the security and surveillance industry. The study emphasizes the importance of accurate gender recognition for applications in biometric verification and user experience enhancement. Our voice recognition system is designed for precise voice gender detection, utilizing advanced Gender Voice Recognition Online technology powered by AI.

Since females exhibit on average a larger acoustic vowel space than males, it is suggested that the stereotype of faster speaking women might arise from this. The female glottis appears to converge more linearly than the male glottis, primarily because of medial surface bulging of the male vocal. This overlap introduces significant complexity, making multi-class gender classification a challenging problem in real-world speech analysis.

Traditional approaches rely on handcrafted feature representations, including Mel-Frequency Cepstral Coefficients (MFCC) and pitch, which capture spectral and prosodic features, such as Linear Predictor Coefficient (LPC). There is a need to go beyond the binary, and to foster more inclusive systems, particularly in speech recognition, speech synthesis, and voice classification, as most existing studies focus primarily on binary classification.

We propose a system that integrates a pre-trained speech representation model, such as Wav2Vec2, to overcome the limitations of traditional approaches. Wav2Vec2 embeddings are high-dimensional, context-aware representations that encode diverse acoustic, phonetic, and linguistic details, and are a self-supervised learning model designed for speech recognition.

Most existing studies focus primarily on binary classification, but there is a need to go beyond the binary and to foster more inclusive systems, particularly for transgender and gender-diverse participants. The study aims to enhance gender and emotion recognition through advanced voice feature extraction, and to evaluate the performance of different speech feature extraction methods, such as MFCC and Wav2Vec2, for multi-class gender classification.

The primary objectives are to evaluate the performance of different speech feature extraction methods, and to determine the impact of feature representation on gender classification performance. Data preprocessing and feature engineering play key roles in data mining initiatives, and machine learning is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data. The study concludes with a discussion on the reproducibility of the results, and highlights the importance of sharing the analysis code and attempting to reproduce one's

own analysis to increase reproducibility.

In this paper, an effective deep-learned and hand-crafted features fusion network is proposed for pedestrian gender recognition, and the impact of combining handcrafted features and deep features for scene image classification is addressed.

II. RELATED WORK

Feature extraction is a critical step in speech recognition, as raw audio signals are complex and difficult for machines to process directly. Mel-Frequency Cepstral Coefficients (MFCCs), introduced by Davis and Mermelstein [1], are one of the most widely used features for speech analysis. MFCCs represent the spectral characteristics of the human vocal tract and are effective for various speech recognition tasks. Other commonly used features include chroma, Mel spectrogram, spectral contrast; however, MFCC and pitch are the primary features used for gender classification in this study.

Pitch represents the fundamental frequency of speech and varies significantly across speakers. Combining MFCC and pitch provides both spectral and prosodic information, improving classification performance in gender recognition tasks [2].

However, traditional handcrafted features have limitations. This limitation is more evident in gender-diverse speech, where voices may not clearly fall into binary categories.

With the advancement of deep learning, more powerful approaches have been developed. Hinton et al. [3] showed that deep neural networks outperform traditional models such as GMMs in speech recognition tasks. Recent models based on self-supervised learning, such as Wav2Vec2, have further improved speech representation learning.

Wav2Vec2 processes raw audio using a convolutional feature extractor followed by a Transformer encoder, enabling it to learn high-level temporal, phonetic, and speaker-specific representations. These learned features are more robust to noise and variability compared to traditional handcrafted features. Recent work by Anidjar and Yozevitch [4] demonstrates that transformer-based approaches can achieve strong performance in gender recognition, particularly in challenging and noisy environments.

This study proposes an approach for identifying speaker gender from audio signals in noisy environments using Wav2Vec2-based embeddings combined with lightweight deep learning models and data augmentation techniques [4]. Such approaches improve the adaptability and inclusivity of modern voice recognition systems by handling diverse speech characteristics.

Gender classification is typically formulated as a supervised learning problem, where models predict the correct label for a given input. Classical machine learning algorithms remain widely used due to their effectiveness and interpretability. Random Forest [5] is capable of modeling complex non-linear relationships and provides insights through feature importance. Support Vector Machines (SVM) [6] identify optimal decision boundaries using kernel transformations in high-dimensional spaces. K-Nearest Neighbors (KNN) [7] is a simple distance-based classifier, while Extreme Gradient Boosting (XGBoost) [8] offers a scalable and high-performance boosting framework.

Recent studies indicate that deep audio embeddings can outperform traditional handcrafted features such as MFCC, as they capture richer and more contextual information from speech signals [4]. However, handcrafted features remain relevant due to their interpretability and lower computational requirements. This creates a need to systematically compare both approaches under a unified framework.

Furthermore, gender-diverse voices often exhibit overlapping acoustic characteristics, which can impact classification performance. Accurate and inclusive gender recognition systems can contribute to improved voice-based applications and better support individuals whose voice characteristics may not align with traditional gender categories. Therefore, this study focuses on evaluating both handcrafted features and deep representations to improve classification robustness and inclusivity.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

Despite substantial progress in speech-based gender recognition, existing systems often struggle with achieving high accuracy and inclusivity across diverse voice categories. Most traditional approaches primarily focus on binary classification (male/female), neglecting transgender voices that exhibit overlapping pitch and spectral characteristics. Furthermore, variations in background noise, audio quality, and speaking conditions introduce inconsistencies that degrade classifier performance.

This paper aims to design and implement a robust and inclusive **multi-class gender classification system** capable of identifying male, female, and transgender voices from Telugu audio recordings. The proposed approach emphasizes effective preprocessing, extraction of meaningful acoustic features such as **MFCCs** and **pitch**, and the incorporation of deep speech representations using **Wav2Vec2** embeddings. By leveraging both handcrafted and learned feature representations, along with data augmentation and comparative analysis, the study seeks to achieve accurate, fair, and scalable gender recognition suitable for real-world audio applications.

B. Objectives

The objectives of this study are as follows:

- 1) **Dataset Collection:** To collect Telugu speech data from male, female, and transgender speakers.
- 2) **Preprocessing and Feature Extraction:** To preprocess the audio data and extract discriminative features, including MFCCs, pitch, and deep speech embeddings using a pretrained Wav2Vec2 model.
- 3) To train and evaluate multiple machine learning classifiers, including Random Forest, SVM, XGBoost, Multi-Layer Perceptron, and KNN for multi-class gender recognition.
- 4) To compare the performance of handcrafted features (MFCC + pitch) with Wav2Vec2-based embeddings using standard evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and confusion matrices, and

identify the most robust approach for inclusive gender classification.

This structured approach ensures a comprehensive evaluation of multiple algorithms under a consistent preprocessing and feature extraction pipeline, while enabling a direct comparison between traditional and deep feature representations for multi-class gender classification using Telugu audio data.

IV. METHODOLOGY

A. Dataset Collection, Preprocessing, and Feature Extraction

The audio dataset was collected from different sources to include a variety of speech samples. To handle class imbalance, data augmentation techniques such as pitch shifting (± 1 semitone), amplitude scaling (± 3 dB), and noise reduction were applied to increase transgender audios. All audio samples were standardized to a 16 kHz sampling rate.

1) *MFCC Feature Extraction*: MFCC features were used to represent the spectral properties of the speech signal by converting it into the frequency domain and applying Mel-scale filters. The MFCC coefficients are calculated as:

$$C_n = \sum_{k=1}^K \log(S_k) \cos \left[\frac{\pi n}{K} (k - 0.5) \right] \quad (1)$$

where S_k represents the energy of the Mel filter bank and C_n denotes the MFCC coefficients. The audio signal was divided into frames of 4000 samples, and 10 MFCC coefficients were extracted from each frame.

2) *Pitch Extraction*: The fundamental frequency (F_0) was computed using the autocorrelation method:

$$F_0 = \frac{1}{T} \quad (2)$$

where T is the time period corresponding to the dominant frequency component. Pitch plays an important role in distinguishing between different voice types.

3) *Feature Fusion*: The extracted MFCC and pitch features were combined into a single feature vector for each audio sample. This combined representation was used as input for the classification models.

4) *Deep Feature Extraction using Wav2Vec2*: In addition to handcrafted features, deep representations were obtained using a pretrained **Wav2Vec2** model. The audio signals were provided as raw waveform inputs after ensuring a sampling rate of 16 kHz.

The model generates contextual embeddings that capture temporal and phonetic information. Since the output embeddings vary in length, mean pooling was applied to obtain a fixed-length **768-dimensional feature vector** for each audio sample. These embeddings were used to compare performance with traditional features.

B. Training, Testing, and Performance Comparison of Classification Models

The dataset was divided into **80% training** and **20% testing** subsets using stratified sampling to maintain class distribution.

Feature scaling was applied using **StandardScaler**, and categorical labels were converted into numerical form using label encoding.

The same training procedure was applied to both MFCC + pitch features and Wav2Vec2 embeddings to ensure a fair comparison.

1) *Random Forest (RF)*: The Random Forest model consisted of **300 decision trees** with a maximum depth of **25**. Class imbalance was handled using **class_weight = balanced**. The final prediction is obtained by averaging the outputs of all trees:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N h_i(x) \quad (3)$$

where $h_i(x)$ represents the prediction of each tree.

2) *Support Vector Machine (SVM)*: The Support Vector Machine model was implemented using an RBF kernel to handle non-linear relationships:

$$f(x) = \sum_{i=1}^n \alpha_i y_i K(x, x_i) + b \quad (4)$$

$$K(x, x_i) = \exp(-\gamma \|x - x_i\|^2) \quad (5)$$

This allows the model to learn complex decision boundaries between different gender classes.

3) *K-Nearest Neighbors (KNN)*: The KNN model used **5 nearest neighbors** for classification. The distance between samples was calculated using Euclidean distance:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (6)$$

Distance-based weighting was applied so that closer neighbors have a higher influence on the prediction.

4) *Multi-Layer Perceptron (MLP)*: The MLP model consisted of two hidden layers with **128 and 64 neurons**. The ReLU activation function was used:

$$f(x) = \max(0, x) \quad (7)$$

The model was trained using the Adam optimizer for a maximum of **300 iterations**, allowing it to learn complex feature relationships.

5) *XGBoost*: The XGBoost model was implemented using gradient boosting. The objective function is defined as:

$$L = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (8)$$

where l represents the loss function and Ω controls model complexity. The model used **250 estimators**, a learning rate of **0.05**, and a maximum depth of **6**.

All models were trained using both MFCC + pitch features and Wav2Vec2 embeddings. MFCC and pitch help capture basic voice features, while Wav2Vec2 provides deeper information from the speech signal.

The results showed that pitch and MFCC features are important for gender classification, whereas Wav2Vec2 gives

a more effective representation and improves performance, especially for transgender voices.

The models were evaluated using Accuracy, Precision, Recall, F1-score, and confusion matrices to measure their performance.

V. EXPERIMENTAL SETUP

This study presents an end-to-end system that extracts features using MFCC and a deep-learning-based approach for gender classification using multiple models. The dataset is designed to isolate and measure purely acoustic effects by synthesizing identical questions with male, female and transgender speakers. The approach is based on standard speech recognition technology by selection of low level features and the design of the system classifies male, female, and transgender voices. Classical features include fundamental frequency and Mel Frequency Cepstral Coefficients (MFCCs). Google Colab is a free cloud-based platform that allows users to write and run Python code using a Jupyter notebook interface, making it easy to perform experiments and visualize results.

A. Dataset Description

Dataset: Telugu Audio Dataset

The dataset contains audio samples of male and female voices, obtained from the OpenSLR Telugu Speech Dataset, and transgender samples were collected from online platforms to ensure diversity. All recordings were standardized to a 16 kHz sampling rate, all audio files were standardized to a uniform sampling rate of 16 kHz and truncated to ensure consistent analysis, all audio was standardized to mono-channel 16 kHz sampling rate for consistent input across models.

The dataset was split into training and testing subsets, with an 80-20 split, the 80–20 split is largely accepted as the standard, it finds a practical compromise between delivering enough data for training and leaving enough examples, typical proportions include a split of 80 percent for training and 20 percent for testing. Table I describes dataset.

TABLE I: Summary of Dataset Characteristics

Attribute	Details
Total Number of Samples	4998 (expanded to 6204)
Male Voice Samples	2,154
Female Voice Samples	2,294
Transgender Voice Samples	532 (increased to 1822)
Sampling Frequency	16 kHz
Audio Duration (Male & Female)	5–15 seconds
Audio Duration (Transgender)	5–15 seconds

B. Model Training and Evaluation

Then there would also be an 80%/20% split between train and test examples, which finds a practical compromise between delivering enough data for training and leaving enough examples.

Split first and then augment data, as data augmentation is used to increase the size of the training set and to get more different data, then enter the train/test split. Cross-validation is a widely used technique in machine learning pipelines, but some of its drawbacks have been recognized in the last

decades, so a neural net with proper cross-validation and bootstrapping always gives better performance.

Pitch and MFCC are extracted from speech signals, and MFCC is a spectral speech representation that reflects the human perception of frequency, using a Mel scale that matches the human ear’s sensitivity. Wav2Vec2 embeddings are high-dimensional, contextualized representations of speech signals produced by the Wav2Vec 2.0 family of self-supervised models, which enables us to build better speech recognition systems for many more languages and domains. To maintain consistency, each track in these databases was segmented into 10-second clips and resampled at 16 kHz, as models typically only work with specific sample rates. The resulting variable-length embeddings were converted into fixed-length feature vectors (768 dimensions) using mean pooling, which treats all elements equally, giving a balanced summary of the sequence and works well when all parts of the sequence are important.

These extracted Wav2Vec2 features were stored as structured datasets and used as input to the machine learning classifiers, ensuring a fair comparison with MFCC+pitch features, and we can leverage Wav2Vec2Processor to process the data to the format expected by the model for training.

Feature scaling was applied using StandardScaler to ensure uniform feature contribution and class-wise behavior was analyzed using Accuracy, Precision, Recall, F1-Score, and confusion matrices, as feature scaling is a critical preprocessing step that standardizes or normalizes data. For models requiring numerical labels (XGBoost and KNN), label encoding was applied.

VI. RESULTS AND ANALYSIS

This work aims to classify human gender based on human voice data between males and females using modified ensemble techniques based on classifiers. This paper proposes a pipeline to create a new model, called Mel and Wave Embeddings for Human Voice Tasks, capable of generating robust embeddings for human voice tasks.

A. Objective 1: Dataset Collection

The dataset was collected from multiple sources, including OpenSLR, which contains male and female recordings of English from various dialects. This dataset was used to create a more balanced representation of all classes during training. It comprises 2,154 male, 2,294 female, and 532 transgender samples, with the transgender class augmented to 1,822 samples (Fig. 1).

By generating synthetic variations of existing data, augmentation ensures the model encounters a more balanced representation of all classes during training. Data augmentation techniques, such as those applied in this study, can achieve significantly improved applicability in tasks involving scarce or imbalanced data.

B. Objective 2: Preprocessing and Feature Extraction

When adopted to the audio domain, data augmentation can be achieved by using some filters on an audio signal such

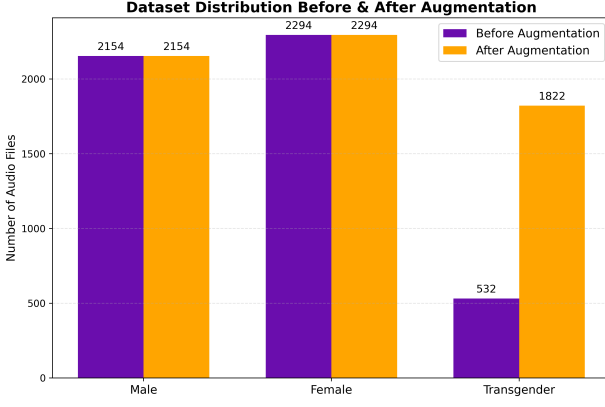


Fig. 1: Sample distribution across gender classes before and after augmentation.

as pitch shifting, removal of noise, and time stretching. Data augmentation for audio involves modifying existing audio samples to create new training data, helping machine learning models generalize better.

After preprocessing, all audio files were standardized to a 16 kHz sampling rate, denoised, and trimmed to remove silence.

The inclusion of augmented samples further enhanced class-wise performance, particularly for transgender voices. Augmentation contributed to higher recall and F1-score by reducing overfitting and improving robustness across diverse acoustic conditions.

The study aims to classify normal and pathological voices by leveraging the wav2vec2 model as a feature extraction method in conjunction with machine learning. Our approach uses wav2vec2 embeddings and mel-frequency cepstral coefficients (MFCC) as input features to a downstream CNN model.

C. Objective 3: Training, Testing, and Model Performance

The **Random Forest** model was trained using MFCC and pitch features to establish a baseline. The dataset was split in an 80:20 ratio for training and testing. The model achieved an accuracy of approximately **93.4%**, with pitch and lower-order MFCC coefficients identified as the most influential features.

In addition to Random Forest, other classifiers including SVM, XGBoost, MLP, and KNN were implemented and evaluated. Each model was tuned using grid search for hyperparameter optimization.

The comparative performance before and after augmentation is presented in Table II.

TABLE II: Performance Comparison of Models Before and After Data Augmentation

Model	Pre-Augmentation (%)				Post-Augmentation (%)			
	Male	Female	Transgender	Overall	Male	Female	Transgender	Overall
Random Forest	98.38	96.97	94.39	97.30	97.68	95.24	97.70	96.78
SVM	96.99	96.54	98.13	96.90	97.45	96.32	99.71	97.66
KNN	89.33	98.26	88.78	93.40	88.17	97.83	98.28	94.60
MLP	97.45	96.97	95.32	97.00	96.75	96.97	98.85	97.42
XGBoost	98.65	94.03	96.12	96.30	98.02	94.75	97.27	96.62

As observed in Table II, the application of data augmentation leads to noticeable improvements in classification performance, with the most significant gains observed in the transgender category across all models.

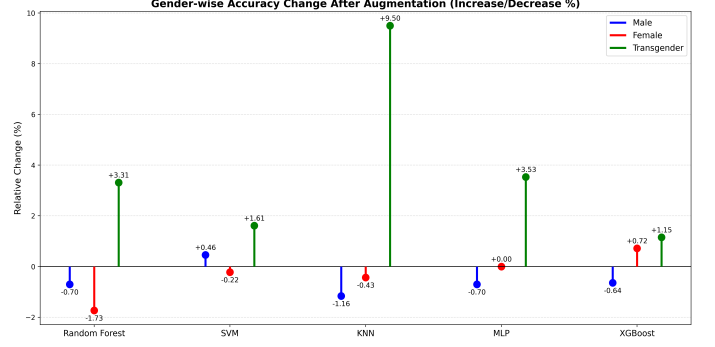


Fig. 2: Percentage change in class-wise accuracy after applying data augmentation.

As illustrated in Fig. 2, transgender classification accuracy increased for all models, with KNN showing the highest improvement.

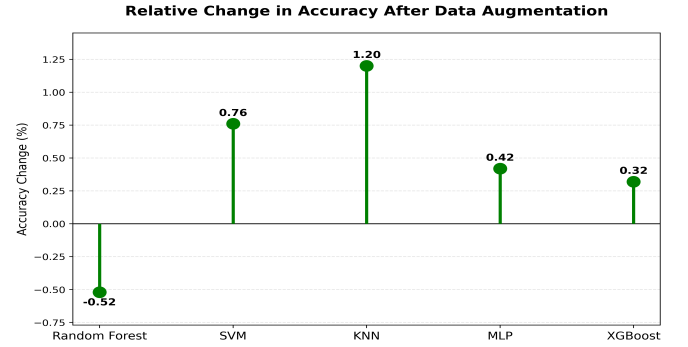


Fig. 3: Variation in overall model accuracy following the application of data augmentation.

As illustrated in Fig. 3, the Random Forest model exhibits a marginal decrease in accuracy after augmentation, likely due to increased variability in the training data, while still maintaining strong performance and interpretability.

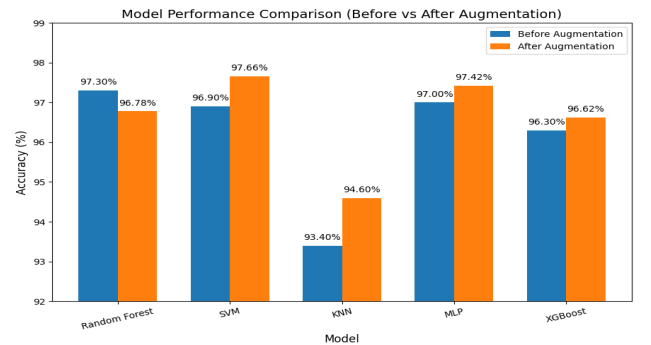


Fig. 4: Comparison of overall model performance before and after data augmentation.

As depicted in Fig. 4, the SVM and MLP models achieve the highest accuracies of 97.6% and 97.4%, respectively, demonstrating strong performance after augmentation.

As illustrated in Fig. 5, Fig. 6, and Fig. 7, pitch and MFCC coefficients consistently emerge as the most influential features

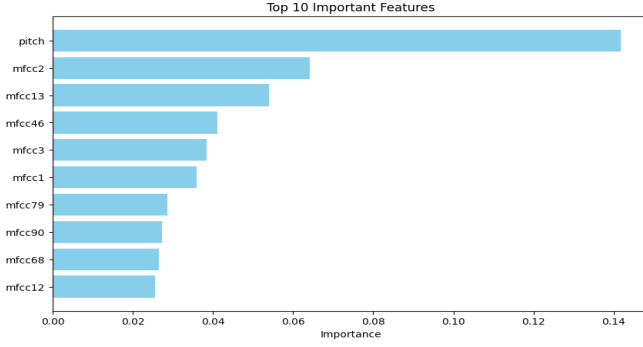


Fig. 5: Top ten features identified by the Random Forest model.

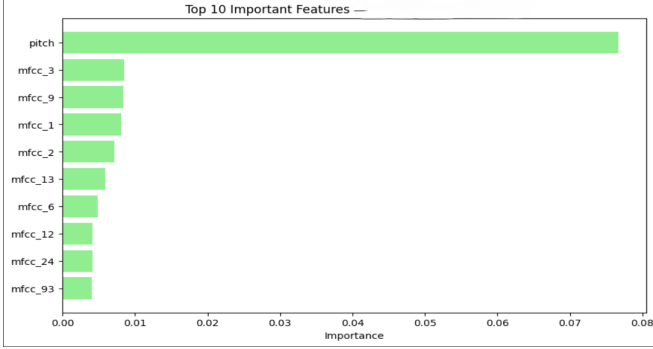


Fig. 6: Top ten most important features contributing to the MLP model.

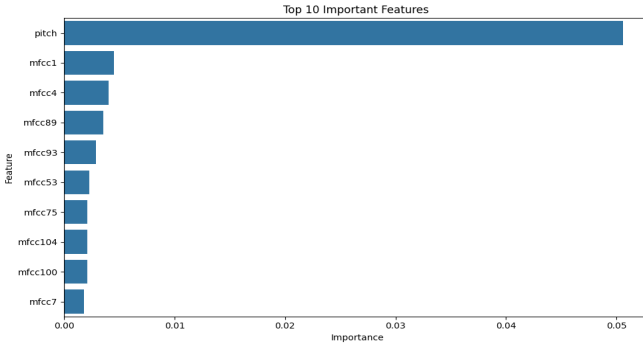


Fig. 7: Top ten dominant features identified by the SVM model.

across all models.

D. Objective 4: Feature Comparison (MFCC vs Wav2Vec2)

The performance of classifiers using Wav2Vec2 embeddings is presented in Table III.

TABLE III: Performance using Wav2Vec2 Embeddings

Model	Accuracy (%)
Random Forest	99.76
KNN	98.97
XGBoost	99.76
MLP	99.84
SVM	99.92

A comparison between MFCC + pitch and Wav2Vec2 features is shown in Table IV.

TABLE IV: Comparison of MFCC + Pitch and Wav2Vec2

Model	MFCC + Pitch (%)	Wav2Vec2 (%)
Random Forest	96.78	99.76
SVM	97.66	99.92
KNN	94.60	98.97
MLP	97.42	99.84
XGBoost	96.62	99.76

As observed in Table IV, Wav2Vec2 embeddings significantly outperform MFCC-based features across all classifiers. This improvement is particularly notable in transgender classification, where deep embeddings capture subtle acoustic variations more effectively.

Overall, the integration of Wav2Vec2 with machine learning classifiers leads to superior performance, demonstrating the effectiveness of deep feature representations for inclusive multi-class gender classification.

VII. CONCLUSION AND FUTURE WORK

This work develops a multi-class gender classification system for Telugu speech that can identify male, female, and transgender voices. MFCC and pitch features, along with data augmentation, helped improve the model's performance and handle class imbalance. Among the models, SVM and MLP performed well, while Random Forest helped in understanding important features. The use of Wav2Vec2 embeddings further improved accuracy by capturing more detailed speech patterns.

The system shows good performance across all gender categories, including transgender voices. Since the features used are not language-specific, the approach can be applied to other languages as well. In the future, the system can be improved by using more data, handling dialect variations, and exploring advanced deep learning models or fine-tuning Wav2Vec2 for better performance.

VIII. INDIVIDUAL CONTRIBUTION

- **Gangabattula Durga Sai Pavan (231IT022):** Dataset collection, Implemented MPL(MFCC + Wav2Vec2 features), Random Forest on Wav2Vec2 features.
- **Kalal Pavan Teja (231IT030):** Implemented data augmentation, KNN (baseline + Wav2Vec2 features), and Random Forest on augmented data.
- **Karella Udayram (231IT032):** Extracted MFCC and Pitch features, implemented XGBoost(baseline + Wav2Vec2) and implemented SVM(Wav2Vec2).
- **Paluru Pavan Kumar (231IT046):** Converted all unique audios to 524 audios (≤ 10 sec) and to standard wav, Tested SVM (MFCC, pitch), Extracted Wav2Vec2 Features for full dataset.

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